Skills Summary

Arithmetic Sequences as Linear Functions

Use this reference while completing your worksheet or when studying.

Skill 1 — A recursive rule from a listed sequence

Two lines, always:

$a_1 =$ first term $\qquad a_n = a_{n-1} + d \quad (n \geq 2)$

Find d by subtracting *consecutive* terms, always later minus earlier, so a decreasing sequence gives a negative d. Check the same difference twice before you commit: one matching gap does not make a sequence arithmetic.

Example: Write the recursive rule for $9, 14, 19, 24, \ldots$ and find a_7.
Step 1. Consecutive differences are constant, so this is arithmetic and the recursive rule is the pair (first term, constant step).
Step 2. $d = 14 - 9 = 5$, so $a_1 = 9$ and $a_n = a_{n-1} + 5$; term 7 is 6 steps along, giving $9 + 6(5) = \mathbf{39}$.

Try it: Write the recursive rule for $40, 33, 26, \ldots$ and find a_6.

check: **5**

Skill 2 — A recursive rule from a situation

Read the context for two things:

the **starting amount** $\rightarrow a_1$ $\qquad$ the **repeated change** $\rightarrow d$

Signal words: "each," "every," "per" mark the step. Gains are positive, losses negative. Decide what a_1 means *before* computing: the amount before any change is a_0, not a_1.

Example: A candle is 21 cm tall and burns down 2 cm each hour. How tall is it after 6 hours?
Step 1. "Each hour" names a constant repeated change, so height is arithmetic in the number of hours, with the burn counted as negative.
Step 2. After n hours the height is $21 - 2n$, so after 6 hours it is $21 - 12 = \mathbf{9}$ cm.

Try it: A plant is 12 cm tall and grows 3 cm each week. How tall is it after 8 weeks?

check: **36**

Skill 3 — Running a recursive rule forward

Count the steps, not the terms:

$a_n = a_1 + (n-1)d$

Between a_1 and a_n there are $n - 1$ arrows, so the step is applied $n - 1$ times. Using n jumps one term too far — that single off-by-one is the most common lost mark on this topic.

Example: $a_1 = 4$ and $a_n = a_{n-1} + 11$. Find a_{10}.
Step 1. Listing ten terms is slow and error-prone; count the applications of the step instead.
Step 2. From term 1 to term 10 is 9 steps, so $a_{10} = 4 + 9(11) = 4 + 99 = \mathbf{103}$.

Try it: $a_1 = 90$ and $a_n = a_{n-1} - 6$. Find a_{12}.

check: **24**

Skill 4 — Recursive rule to explicit linear function

The step is the slope:

$a_n = a_1 + (n-1)d = dn + (a_1 - d)$

So the slope is d and the value at $n = 0$ is $a_1 - d$ — one step *back* from the first term. Writing a_1 as the intercept is the classic error; check it by testing $n = 1$.

Example: $a_1 = 8$ and $a_n = a_{n-1} + 5$. Write the explicit rule and find a_9.
Step 1. A constant step means the sequence is a linear function of n, so rewrite the rule as slope times n plus an intercept.
Step 2. $a_n = 8 + 5(n-1) = 5n + 3$, so $a_9 = 5(9) + 3 = \mathbf{48}$.

Try it: $a_1 = -2$ and $a_n = a_{n-1} + 7$. Write the explicit rule and find a_{11}.

check: **68**

(!) Watch out: A recursive rule without its first line is not a formula — $a_n = a_{n-1} + 5$ fits every sequence that counts by fives. Always write a_1, and always state the domain $n \geq 2$ on the step line.